

PVPIT Grand Hackathon 3.0

- **Problem Statement Title-** GazeType - Eye tracking keyboard for unfortunate people with disabilities
- **Theme-** Human Assistive Technology and Human - Computer interaction
- **PS Category-** Software
- **Team Name -** Byte Builders
- **Team Members -** Parth Verma, Anushka Wankhede, Nupur Talathi, Vedika Kulkarni, Shroojan Dhok, Ayush Taske



INTRODUCTION

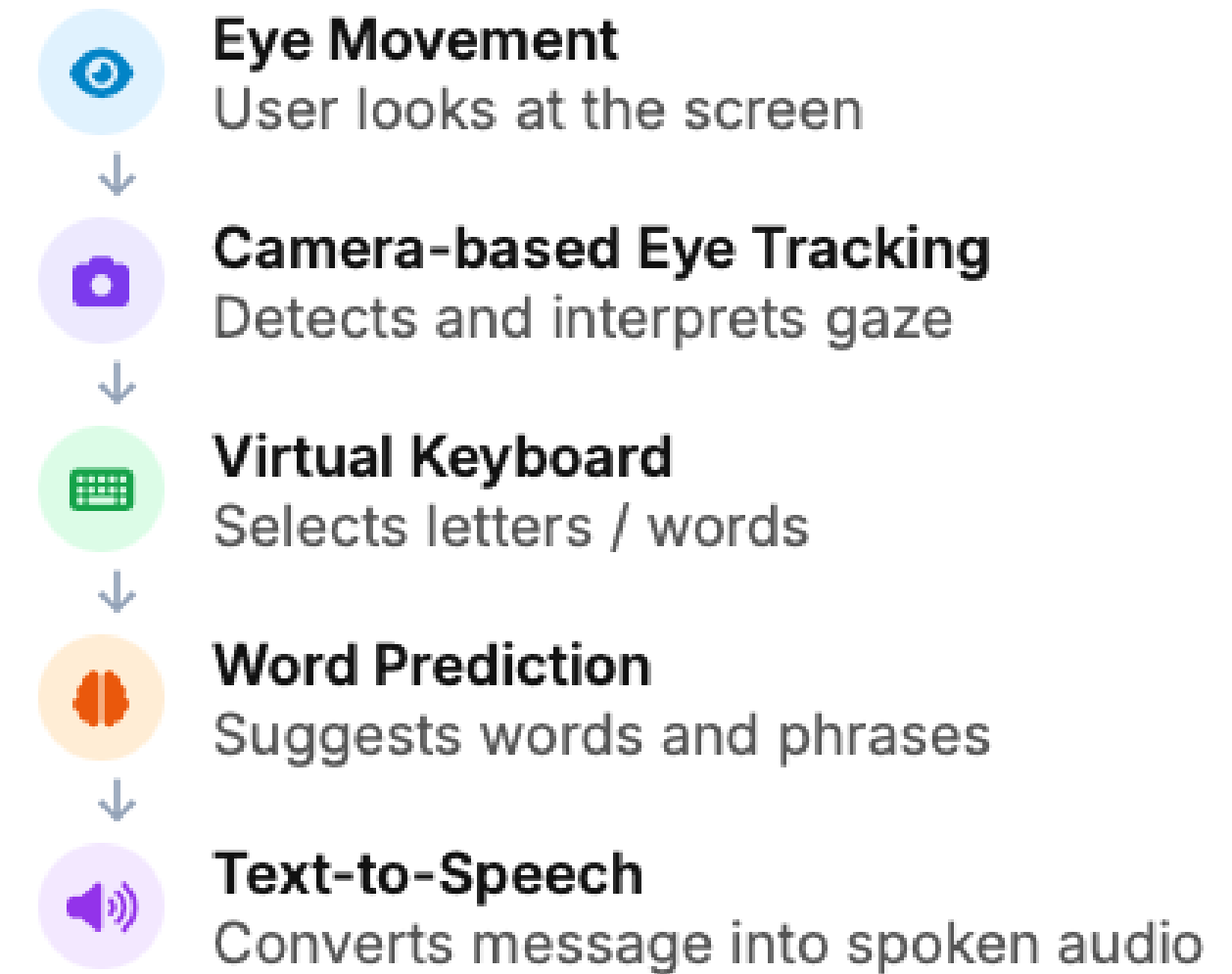
THE CHALLENGE:

People with severe motor disabilities may lose the ability to speak or use conventional input devices, making it difficult to express their needs, thoughts and emergencies. For some users, eye movement may remain a usable channel for communication



OUR SOLUTION:

A Camera based eye-gaze communication system that converts controlled eye movements into text and speech



When speech and physical movement are no longer reliable ways to communicate, even a simple **“I need help”** can become difficult to express. By transforming **eye movement into text and speech**, our system aims to give users a faster, more independent way to communicate their needs, thoughts and emergencies.



Speech
(lost/limited)

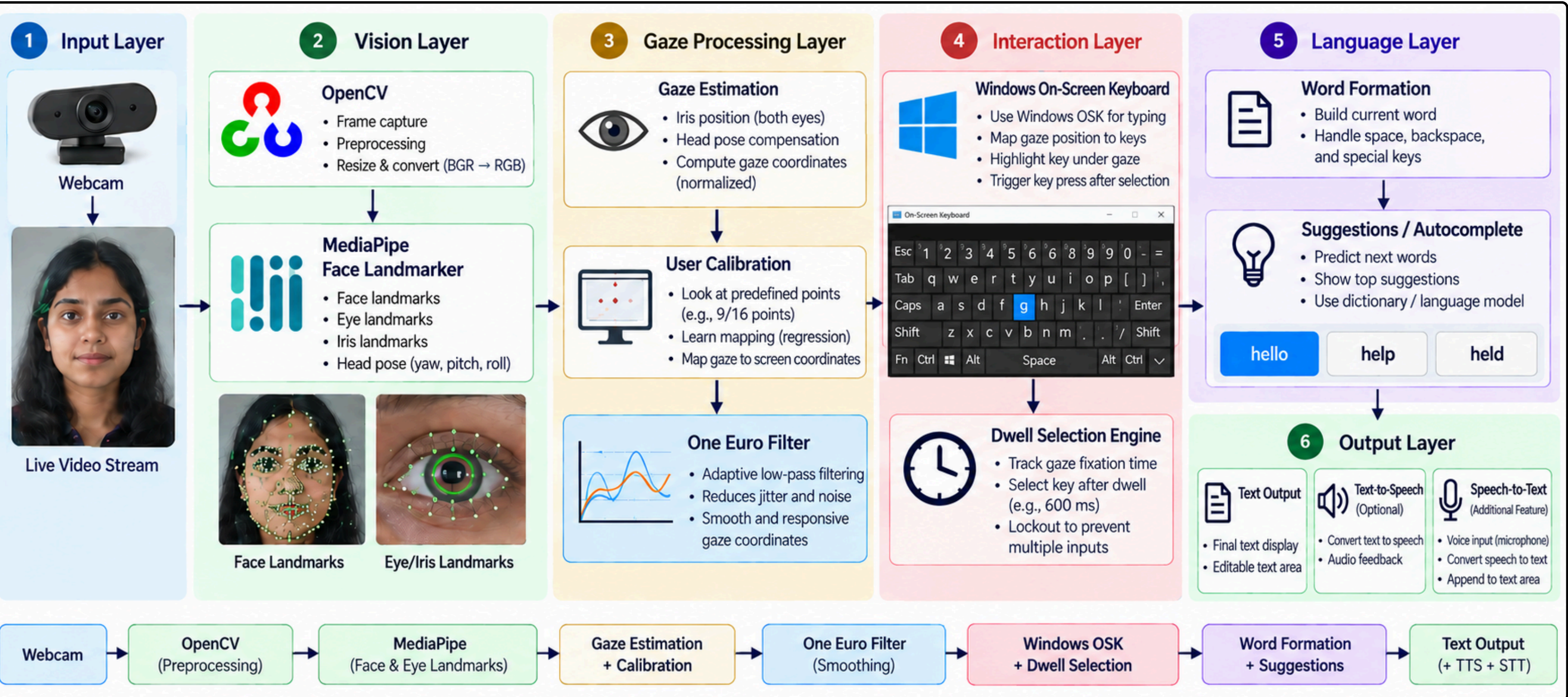


Physical Input
(not possible)



Eye Movement
(still possible)

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

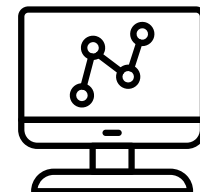
Built with existing technology, designed for low-cost accessibility, and scalable from a laptop prototype to a dedicated assistive device

ANALYSIS OF THE FEASIBILITY OF THE IDEA



Existing Eye-Tracking Technology

Webcam-based and infrared eye-tracking techniques can track gaze direction without requiring physical input.



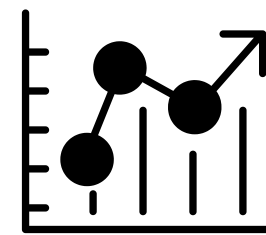
Software-Based Implementation

The system can be developed using computer vision, gaze estimation, and machine learning, reducing the need for complex hardware.



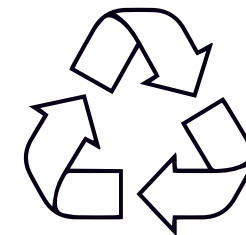
Simple User Interface

A virtual keyboard with large, well-spaced keys can be designed specifically for users with limited movement.



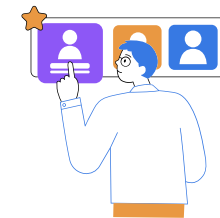
Scalable Architecture

The prototype can start with a webcam and laptop, and later be upgraded to dedicated eye-tracking hardware for improved accuracy.



Reusable Across Conditions

The same system can assist people with different severe motor or speech impairments rather than being limited to one condition.



Personalization Increases Usability

Custom keyboards, frequently used phrases, language selection, and personalized vocabulary can make the system more practical for individual users.

POTENTIAL CHALLENGES AND RISKS



1. Lighting Variability

Glare, low light, or backlighting degrades iris and pupil detection, making gaze estimates unreliable. Real-world environments rarely match the controlled lab



2. Calibration drift

Head shifts or posture changes during a session slowly throw off the gaze-to-screen mapping.



3. Diverse user conditions

Nystagmus or involuntary eye movement can reduce accuracy for some of the very users the system targets.



4. Privacy concerns

Continuous webcam capture of the face/eyes is sensitive data,

MITIGATIONS



Adaptive preprocessing

Auto-exposure/histogram correction in OpenCV, plus optional IR webcam support.



Personalized profiles

Per-user calibration with fallback modes like blink or switch-based selection.



Continuous recalibration

Head-pose compensation with periodic auto re-calibration during long sessions.



On-device processing

Video stays local; no raw frames stored or transmitted.

Your Team
Name

IMPACT AND BENEFITS

They may lose the ability to speak, but they should never lose the ability to be heard



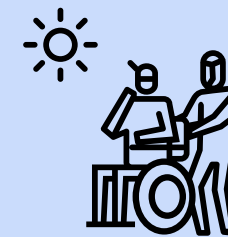
Gives a Voice to the Voiceless

Enables people who cannot speak to express their thoughts, needs, and feelings.



Enables Immediate Communication of Critical Needs

Allows users to communicate pain, discomfort, emergencies, or urgent requests



Eliminates Guesswork for Caregivers

Helps caregivers understand what the person actually wants instead of interpreting unclear gestures or expressions.



Restores Personal Independence

Gives individuals the ability to make choices and communicate decisions without constantly depending on another person.

RESEARCH AND REFERENCES

1. Existing Research Papers

	Paper (Year)	Main Contribution	Limitation
1	Karatay et al. (2024) <i>Real-time eye movement-based computer interface</i>	Webcam-based, software-only gaze-controlled virtual keyboard for real-time interaction.	Basic communication; scope for better prediction and personalization.
2	Nguyen et al. (2023) <i>On-screen keyboard controlled by gaze</i>	Designed gaze keyboard specifically for ALS users.	Language/layout specific; traditional gaze typing remains slow.
3	Jeevithashree et al. (2021) <i>Eye gaze controlled adaptive virtual keyboard</i>	Adaptive keyboard for severe speech/motor impairment.	Ocular conditions like nystagmus or strabismus create interaction challenges.
4	Cai et al. (2024) <i>SpeakFaster</i>	LLM-based word/phrase prediction for eye-gaze AAC.	Tested with only 2 eye-gaze ALS users; personalization remains a future direction.

2. Existing Solutions Today



Tobii Dynavox

- Dedicated eye-tracking AAC devices.
- Supports dwell/switch selection.



Eye-gaze virtual keyboards

- Select letters using gaze/dwell.
- Works on computers/tablets.



AAC communication software

- Text-to-speech, symbol-based communication.
- Helps with daily interaction.



LLM-based systems (e.g., SpeakFaster)

- Predicts next words/phrases to reduce keystrokes.
- Early-stage research and limited real-world use.

3. Limitations of Existing Solutions

- **Expensive hardware** — Dedicated eye-tracking devices are costly.
- **Slow typing** — Gaze typing is time-consuming.
- **Letter-by-letter input** — No smart suggestions or autofill.
- **Generic prediction** — Not personalized to user's vocabulary and needs.
- **Complex setup** — Calibration and configuration can be difficult.
- **Limited functionality** — Mostly focused on communication, not full computer interaction.

4. Existing Solutions vs Our Solution

Feature / Aspect	Existing Solutions	Our Solution
Hardware needed	Dedicated eye-tracking device (costly)	Normal webcam (low-cost)
Typing speed	Slow (letter-by-letter)	Faster with word suggestions
Input method	Letter selection	Letters + autofill + common phrases
Prediction	Generic / limited	Personalized vocabulary (ML/NLP based)
Setup & calibration	Complex	Simple & easy
Functionality	Mostly communication	Communication + basic computer interaction
Cost	High	Low

Thank You!

-Team Byte Builders

